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DAP Asymchem Sandwich, UK. (INCOTERMS2010)

NO.	Model/Description	Technical Configuration	QTY	Options	Remark	Unit Price (£)	Optional Price (£)	Machine Price (£)
A-Freeze Dryer		UKCA, CE Complied						
A-1	LYO-5(SIP,CIP)-RBR Type							
A-1.1	Freeze Dryer-LYO-5 (SIP/CIP) -RBR Type-Basic Machine	Consists of: Chamber; Shelf unit; Main Door & Pizza Door; Condenser; Main isolation valve; Refrigeration system; Circulation system; Vacuum system; Aeration system; Hydraulic system; Pneumatic system; CIP/SIP system; Control system; Standard documentation;	1		1.CE standard;PED vessel 2.Pneumatic piping material:SS 304	440,000		440,000
A-1.2	Calibration certificates of the instrument	A set of Instrument calibration certificates Calibration certificates are valid only for the duration of FAT, it don't include spare parts.	1					Included
A-1.3	Overpass, stair, guardrail	A set of overpass, stair and guardrail to reach the top of the chamber&condenser for instruments calibration and component maintenance.	1					Included
A-1.4	Water Flow Control Valves for Each Compressor	Add water flow control valves for each compressor, total two sets.	1					Included
A-1.5	Modulated Vacuum Control	★Using proportional leakage control system to replace the manually valve system, the chamber pressure control will become more precise. ★Key Features: 1. Aerated gas process is more stable, vacuum control is more precise, and the quality of the freeze drying product is less affected. 2. In the production of different products, the lyophilization process is different, compared with the conventional manual valve, automatic adjustment can be realized, reducing the dependence on personnel. ★Type: Digital Mass Flow Controller ★Brand :Azbil ★Number :One(1)>One(1)	1					Included
A-1.6	Cooling heat exchange for circulation system	In the circulation system, a heat exchanger is connected in series, and the cooling water is used to exchange heat with the silicone oil, thereby heat exchange to the shelves and then radiated to the chamber. ★Key Features: 1.Reduce the cooling time after SIP. Cooling time:Depends on the temperature & flow of the cooling water 2.Reduce the damage to the compressor when the compressor is used to cool the silicone oil ★Special design by Tofflon ★Material: SS 304 Utility Requirements (Approx.): Cooling water: 20m3/h @ 1.5~3bar, 15~30°C	1					Included
A-1.7	Dry Pump System Instead of Oil Pump System	Dry vacuum pump to eliminate the possibility of oil back streaming from rotary pumps and reduce maintenance time. Dry vacuum brand : Edwards ★Model: GV80 ★Number: One(1) ★Motor power: 4kW /pcs Roots pump Brand: Edwards ★Model: EH250 ★Number: One(1) ★Motor power: 2.2kW /pcs Utility Requirements (Approx.): ★Cooling water for vacuum system: 3L/min @ 2~3bar, 15~30°C (Recommended temperature: 25°C) ★Dry nitrogen gas consumption for vacuum system: 20L/min @ 4.5bar Cooling water inlet valve ★Type: Solenoid valve ★Number: One(1)	1					Included
A-1.8	RAID1 function for PC	★There is RAID1 function for PC. ★Brand: Samsung or equivalent	1					Included
A-1.9	Pressure transmitter on compressor	The compressor is equipped with pressure transmitters for high, medium, low, and oil pressure to monitor operating conditions. All pressures can be displayed on the SCADA system. ★Number: Eight(8)	1					Included
A-1.10	Absolute Pressure Measurement on Chamber	Add a capacitance vacuum gauge installed on the chamber, Key Features: Compared to resistive vacuum gauge (Pirani), its measurement result are not affected by gaseous media, which can provide greater vacuum measurement accuracy. ★Brand :MKS ★Range : 0-1mbar ★Accuracy: 0.5%	1					Included
A-1.11	Jacket cooling water circulation	Cooling water in the jacket can be recirculated through the utility piping system rather than discharged, to reduce water consumption.	1					Included
A-1.12	BAG-IN/BAG-OUT Filter at Vacuum Pump Outlet	There is one BAG-IN/BAG-OUT (BIBO) Filter at vacuum pump outlet. ★Brand: Camfil ★Number: One(1) Note: That the pipelines between vacuum pump outlet and BIBO isn't in Tofflon's scope of supply. Customer shall provide and connect pipelines.	1					Included
A-1.13	Conductivity sensor	A conductivity sensor is installed on CIP drain line, to measure the conductivity of the wastewater. ★Number: One(1)	1					Included
A-1.14	Condenser hot water defrosting	The condenser supports hot water defrosting. Overflow vessel connections and relevant control valves are provided.	1					Included
A-1.15	External CIP System with Two Tanks One Pump	Including two external CIP tanks, a centrifugal pump, and associated control valves. Key Features: 1.It is designed to supply sufficient flow and pressure head to the set of nozzles and spray balls working simultaneously. 2.Located in an ATEX Zone; all relevant components, including pump, sensors, cables, are selected as Ex Type. 3.No dedicated electrical cabinet; the system shares the electrical cabinet and control system with the freeze dryer, and all operations can be performed via the freeze dryer control system. External CIP Tank: ★PW tank: Volume: 2000L; Number: One (1); Material: SS 316L; Internal Finish: Ra≤0.4µm; Cleaning water inlet/outlet valve: Included; Sterilization method:80℃ pasteurization; Spare Port:TC 2" *1 ★WFI tank: Volume: 2000L; Number: One (1); Material: SS 316L; Internal Finish: Ra≤0.4µm; Cleaning water inlet/outlet valve: Included; Sterilization method:80℃ pasteurization; Spare Port:TC 2" *1 Sanitary centrifugal pump:★Brand: Fristam; ★Number: One (1) Process control valves: Included Filter:★Core aperture: 0.22µm Tank washing:★Sprayer: Included Sanitary pipe:★Material: SS 316L; ★Welding: Orbitally welded Liquid level sensor:★Brand: ENBBON Temperature sensor at outlet: Included Water flow controller:★Brand: IFM Pressure meter: Included Electrical cabinet: Included Related software functions: Included Maintenance ladder and guardrail: Included Steam heat exchanger:★Type: Sanitary heat exchanger; ★Capacity: capable of heating a tank from ambient temperature to 80°C within 1 hour (by recirculation); ★Number: One (1); ★Vessel standard:PED	1					Included
A-2	Pure steam generation system							
A-2.1	Pure steam generation system PSG-500-S	300 kg / h, 6-8bar plant steam	1			60,000		60,000
						Sub Total for A		500,000
B-Loading & Unloading System		UKCA, CE Complied						
B-1	Slurry Loading System	Including: Peristaltic pump Balance						
B-1.1	Slurry Loading System		1		Including PE bag support, PTFE shovel	95,000		95,000
B-2	Containment Isolator							

B-2.1	Negative Pressure Containment Isolator for Dispensing (Isolator-T-01)	1.Passbox chamber(1 glove) + Tray Storage Chamber(3 gloves)+Loading&Unloading Chamber(5 gloves)+Powder Collecting Chamber(3 gloves)+Used trays Storage and Washing Chamber(6 gloves);, Cladding boards, Ducts and Frameworks 2. Air Filtration Unit (PUSH-PUSH) 3. Glass Panel / Doors/ Glove ports 4. Cleaning & Draining System 5. Electrical Control System 6. Instruments & Gauges 7.Slope baseplate design(300*10) 8.350Lux White Light(LED) 9.0.5%vol/hr@100Pa leakage rate 10.UPS for PLC 30min control 11.Humidity&temperature sensor 12.Nitrogen Mode 13.CE certification	1		Equipment Dimensions:L*W*H unit:mm Passbox:650*800*2600 Trays Storage Chamber:1275*800*2600 Loading&Unloading Chamber:3200*800*2600 Powder Collecting Chamber:1500*800*2600 Used trays Storage and Washing Chamber:2300*800*2600 1.Non-sterile environment in the chamber; 2.The electronic scale is not included in this supply scope. 3.The Isolator meets the requirements for operating QIBS products, and customers can conduct SMEPAC testing through a third party 4.Excluding Oxygen concentration sensor.	100,000		100,000
B-2.2	Continuous Liner for Isolator	Tofflon/DN250: UPE	1		DN250.6m-UNFOLDED BAG			Included
B-2.3	Continuous Liner for Isolator	Tofflon/DN350: UPE	1		DN350.6m-UNFOLDED BAG			Included
B-2.4	Dedicated Rotary Dehumidifier Unit for Isolator	Tofflon 1. Baseplate + Insulation Cavity 2. Cooling / Dehumidifier 3. Heating + Humidifier 4. Control System (IEC) + Utilities Valves	1					Included
B-2.5	Oxygen sensor	TEREN G3Ex;	5		Measuring range between 0 and 25% with a 1%VOL accuracy. Customised alarm and interlock strategies for oxygen concentration			Included
B-2.6	Wireless Glove Integrity Testing System	Tofflon	1		Integrated with Isolator.			Included
B-3	ALUS for Trays							
B-3.1	Automatic loading unloading system	including: Pushing system; Unloading system; Docking system; Control system; Beltcon	1		Including CE Certificate	204,000		204,000
Sub Total for B								399,000
C-Documentation (Execution is not included)								
C-1	Design Qualification(DQ) Protocol		1		For whole line			Included
C-2	Factory Acceptance Test (FAT) Protocol		1		For whole line			Included
C-3	Site Acceptance Test (SAT) Protocol		1		For whole line			Included
C-4	Installation Qualification(IO) Protocol Operational Qualification (OQ) Protocol		1		For whole line			Included
C-5	Function Specification(FS) Design Specification(DS) Hardware Design Specification(HDS) Software Design Specification(SDS) Validation Plan(VP) Risk Assessment(RA) Requirements Traceability Matrix(RTM) Validation Summary Report(VSR)		1		For whole line			Included
C-6	Instrument Calibration Certificates	Third Party Calibration Before FAT	1		For whole line			Included
Sub Total for B						0		34,000
Sub Total for (A+B+C)						0		933,000
D-Service								
D-1	Factory Acceptance Test (FAT)	Based on FAT protocol, 2~3 persons from customer to Tofflon about 7 days for FAT, travel cost will cover by customer.	1		On completion of assembly of the equipment and internal checks and tests, the equipment can be tested in the presence of the client or their representative prior to shipment. The test protocol will be provided to the client prior to performing FAT and, on satisfactory completion, the protocol is signed by both parties.			Included
D-2	Installation and commissioning	3 engineers from Tofflon will take about 1 days in customer plant for Unpacking , accommodation in the customer place and travel cost will bear by the buyer.	1		Unpacking (Customer scope Tofflon assists)			Included
		3 engineers from Tofflon will take about 2 days in customer plant for Machine Postioing, accommodation in the customer place and travel cost will bear by the buyer.	1		Machine Postioing (Customer scope Tofflon assists)			Included
		3 engineers from Tofflon will take about 10 days in customer plant for Restore to install , accommodation in the customer place and travel cost will bear by the buyer.	1		Restore to install the machines			Included
		3 engineers from Tofflon will take about 7 days in customer plant for Commissioning, accommodation in the customer place and travel cost will bear by the buyer.	1		Commissioning (All the utilities to be provide by Customer)			Included
D-3	Site Acceptance Test (SAT)	Based on SAT protocol. Besides labour and services associated with undertaking the standard SAT. 4 engineers from Tofflon will take about 10 days in customer plant for SAT, accommodation in the customer place and travel cost will bear by the buyer.	1		- The client is responsible for offloading, unpacking, positioning, connection of utilities and ensuring the required utilities are available . - Tofflon technicians or its nominated representative will reit items disassembled from the system for shipment, recharge all fluids and undertake the initial adjustment of the system. - On completion of the installation phase Tofflon will, in the presence of the client, undertake a standard acceptance protocol (SAT) to ensure the system performs satisfactorily. The SAT is signed by both parties on satisfactory completion. - SAT undertaking will be within six (6) months after delivery.			Included
D-4	Training	2 Tofflon engineers Maintenance and Operation training for 3 Days in customer plant,accommodation in the customer place and travel cost will bear by the buyer.	1					Included
D-5	IQ & OQ Execution	Based on IQ/OQ protocol. Besides labour and services associated with undertaking the standard IQ/OQ 4 engineers from Tofflon will take about 10 days in customer plant for IQ/OQ, accommodation in the customer place will bear by buyer.	1		Accommodation in the customer place will bear by buyer.			Included
D-6	Packing		1					Included
D-7	Transportation	DAP Asymchem Sandwich, UK. (INCOTERMS2010) including insurance	1			33,750		33,750
sub total D								33,750
Total(A+B+C) +D								966,750
Commercial terms								
1	Payment Terms	*30% advance by T/T along with Purchase Order *60% of the payment is due via T/T within 5 days after the completion of the Factory Acceptance Testing (FAT). *10% of the payment is due via T/T within 5 days upon receipt of Shipment Notice from the Seller						
2	Delivery	To be FAT within 8-10 months upon receipt of technically and commercially clear sales contract along with advance payment. The shipment from Shanghai, China will depends on receipt of export license if require.						
3	Guarantee	The equipment is warranted for 12 months after completion of SAT or 18 months after the date of despatch from Tofflon workshop, whichever is the earlier. The guarantee applies only to the replacement or rectification of faults arising from poor manufacturing workmanship.						
4	General Conditions	Accord to Tofflon GCC (Attached)						
5	Validity	60 days						